MagGeo - Annotate geo-magnetic satellite data with GPS trajectories

Data-fusion tool to combine geo-magnetic data with GPS trajectories

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What's MagGeo

MagGeo is a tool that helps ecologists or animal movement researchers to link earth's magnetic field data from satellite source to GPS trajectories. Inspired by the Environmental Data Automated Track Annotation System (Env-DATA) Service a tool from Movebank and help researcher to get a better understanding about the geomagnetic variations across the GPS trajectories.

MagGeo is entirely built-in python and includes a two Jupyter Notebooks that offer two ways to link GPS tracks with the geomagnetic components using the data from one of the up-to-date satellite sources - Swarm Constellation. MagGeo will create an enriched GPS track with the following components:

- **Latitude** from the GPS Track.
- **Longitude** from the GPS Track.
- **Timestamp** from the GPS Track.
- **Magnetic Field Intensity** mapped as Fgps in nanoTeslas (nT).
- **N (Northwards) component** mapped as N in nanoTeslas (nT).
- **E (Eastwards) component** mapped as E. in nanoteslas (nT).
- **C (Downwards or Center)** component mapped as C in nanoTeslas (nT).
- **Horizontal component** mapped as H in nanoTeslas (nT).
- **Magnetic Declination or dip angle** mapped as D in degrees
- **Magnetic Inclination** mapped as I in degrees
- **Kp Index** mapped as kp
- **Total Points** as the amount of Swarm measures included in the ST-IDW process from the trajectories requested in the three satellites.
- **Minimum Distance** mapped as MinDist, representing the minimum distance amount the set of identified point inside the Space Time cylinder and each GPS point location.
- **Average Distance** mapped as AvDist, representing the average distance amount the set of distances between the identified Swarm Point in the Space Time cylinder and the GPS Points location.

Researchers, particularly ecologists now can study the annotated table to analyze the geomagnetic Spatio-temporal variation across any GPS trajectory.

Suggestions?

MagGeo is work in progress and we are constantly making improvements that you can follow up with the commits made in the pubic GitHub repo. For general inquiries, scientific concepts, suggestions please email: Fernando.Benitez@st-andrews.ac.uk, ud2@st-andrews.ac.uk, jed.long@uwo.ca

For **errors**, or **improvements** please submit an issue in this repo, describing the problem.

About MagGeo

Inspired by The Environmental Data Automated Track Annotation System (Env-DATA) Service a tool on Movebank, where ecologists and animal movement researchers all over the world can link movement data with global environmental datasets. Including hundreds of variables from a diverse set of data sources including the European Space Agency (ESA), the National Aeronautics and Space Administration (NASA), the US National Oceanic and Atmospheric Administration (NOAA), and others. EnvData allow researchers to annotate in space and time multiples environmental information to enrich their GPS tracks to analyze the influence of several environmental variables in the trajectory. Using the Env-DATA Track Annotation Service registered users on MoveBank are able to get environmental parameters—such as wind conditions, land use, vegetation, and snow cover—for the whole world. Using different interpolation methods users can include multiple environment variables selecting from a comprehensive list of datasets (you can browse the available dataset here).

The second element that inspired **MagGeo** is having a tool to help researcher to get a better understanding over how the earth's magnetic field is being used by birds as one of their navigational strategies. Despite of there are several approach in this regards we know still have little knowledge about how birds can use the influence of the magnetic field for their migration patterns, especially for those long-distance migratory animals. Other studies have been reflecting into the magnetic field influence based on magnetic field estimation models, or using some displacement experiments with particular species. The disadvantage of those previous studies is the magnetic field is a highly dynamic force that have different impact around the earth every day. Having said that MagGeo wants to take advantage of what is considered best survey of the geomagnetic field and its temporal evolution - Swarm Constellation. Swarm is a ESA's magnetic field mission, launched on 22 November 2013, consists of the three identical **Swarm satellites** (**A**lpha, **B**ravo, and **C**harlie). Swarm A and C flying side-by-side (1.4° separation in longitude) at an altitude of 462 km (initial altitude) and Swarm B at higher orbit of 511 km (initial altitude) are equipped with the following set of identical instruments.

The data products available from Swarm are Level 1b and Swarm Level 2 products. These products include Swarm magnetic field models, ionospheric and thermospheric products, and others. MagGeo use the Swarm Level 1b data product as the corrected and formatted output from each of the three Swarm satellites. For more information about the Swarm Data Products click here.

MagGeo has been deployed using a set of Jupyter notebooks a powerful tool to run a python environment. Completely build in python 3.8 MagGeo is a well described program that will

guide you through several steps to annotate your GPS trajectories with the geomagnetic field components reported by Swarm. You can access to Swarm Data products via HTTP or FTP using :

- via any HTTP browser at http://swarm-diss.eo.esa.int
- directly via an ftp client at ftp://swarm-diss.eo.esa.int

However **MagGeo** use **VirES** (Virtual environments for Earth Scientists) a platform for data & model access, analysis, and visualisation for ESA's magnetic mission **Swarm**. This is a powerful client with the viresclient API that provide several classes and methods defined in the vires client package. The `viresclient` Python package allows you to connect to the VirES server to download Swarm data and data calculated using magnetic models.

Part I

How to use it

Installing MagGeo

MagGeo is a specialized tool that helps researchers analyze magnetic field data in relation to animal movement patterns. This guide will walk you through installing MagGeo on your computer, even if you're more familiar with R than Python.

What You'll Need

Before we begin, understand that **MagGeo** is currently built using Python (similar to how some R packages depend on specific R versions). We'll set up a contained environment so it won't interfere with any existing R installations or other software on your computer.

Step 1: Install Miniconda (Your Python Environment Manager)

What is Miniconda? Think of Miniconda as a package manager for Python, similar to how CRAN manages R packages. It creates isolated environments so different projects don't conflict with each other.

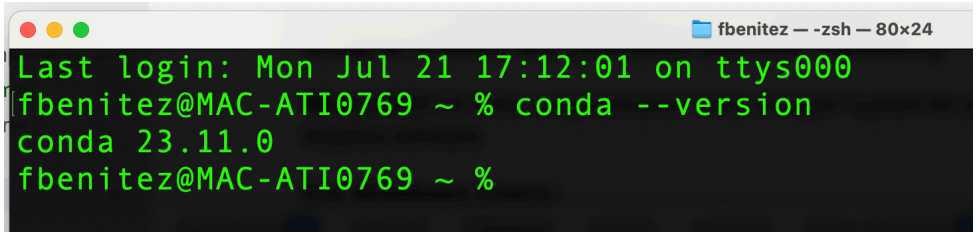
For Windows Users:

1. Visit https://docs.conda.io/en/latest/miniconda.html
2. Download the "Miniconda3 Windows 64-bit" installer
3. Run the downloaded `.exe` file as administrator
4. Follow the installation wizard:
   - Accept the license agreement
   - Choose "Just Me" installation
   - Use the default installation location
   - **Important:** Check "Add Miniconda3 to my PATH environment variable" (even if it shows a warning)
5. Click "Install" and wait for completion

For macOS Users:

1. Visit https://docs.conda.io/en/latest/miniconda.html
2. Download the "Miniconda3 macOS 64-bit pkg" installer
3. Double-click the downloaded `.pkg` file
4. Follow the installation wizard using default settings
5. The installer will automatically add conda to your PATH

Verification: After installation, open a new terminal (macOS) or Command Prompt (Windows) and type `conda --version`. You should see a version number.



Step 2: Clone MagGeo

You will need to clone the MagGeo repository in your local computer (we are currently working to make make MagGeo available in `pip` to an easier installation), for now simply clone it and you will have access to all MagGeo.

If you don't have **Git**, or you are not use to work with version control, you will need to install git to clone any repository from GitHub

Installing Git

What is Git? Git is like a sophisticated "track changes" system for code, similar to version control in collaborative documents. We need it to download the MagGeo software.

For Windows Users:

1. Visit https://git-scm.com/downloads
2. Download "Git for Windows"
3. Run the installer with these recommended settings:
   - Select "Use Git from the Windows Command Prompt"
   - Choose "Checkout Windows-style, commit Unix-style line endings"
   - Use all other default settings

For macOS Users:

Option A (Recommended):
1. Open Terminal (found in Applications > Utilities)
2. Type `git --version` and press Enter
3. If **Git** isn't installed, macOS will prompt you to install Xcode Command Line Tools
4. Click "Install" and follow the prompts

Option B:
1. Visit https://git-scm.com/downloads
2. Download "Git for macOS"
3. Run the installer with default settings

For Both Windows and macOS Users:

1. **Open your command line interface:**
   - **Windows:** Press `Windows + R`, type `cmd`, press Enter
   - **macOS:** Press `Cmd + Space`, type "Terminal", press Enter
2. **Navigate to your desired location** (optional but recommended):

```
# Windows users:
cd C:\Users\YourUsername\Documents

# macOS users:
cd ~/Documents
```

3. **Download MagGeo:**

```
git clone https://github.com/MagGeo/MagGeo-Annotation-Program.git
```

   This creates a folder called "**MagGeo-Annotation-Program**" with all the necessary files.

Step 4: Navigate to the MagGeo Folder

```
cd MagGeo-Annotation-Program
```

Step 5: Create the MagGeo Environment

This creates a specialized Python environment with all the specific packages MagGeo needs, similar to installing a comprehensive R package with all its dependencies.

```
conda env create --file environment.yml
```

> ⚠ Warning
>
> **This process will take 5-10 minutes** as it downloads and installs all required components. You'll see progress messages - this is normal.

Step 6: Activate the MagGeo Environment

Every time you want to use MagGeo, you'll need to "activate" its environment first:

```
conda activate MagGeoEnv
```

What this does: Switches your command line to use the MagGeo-specific Python environment, ensuring all the right tools are available.

Step 7: Set Up VirES Access (Magnetic Field Data)

What is VirES? VirES provides access to satellite magnetic field data from ESA's Swarm mission. Think of it as a specialized database for magnetic field measurements that MagGeo uses to understand the magnetic environment your animals experienced.

MagGeo use **VirES** (Virtual environments for Earth Scientists) a platform for data & model access, analysis, and visualization for ESA's magnetic mission **Swarm**. This is a powerful client with the viresclient API that provide several classes and methods defined in the vires client package. The `viresclient` Python package allows you to connect to the VirES server to download Swarm data and data calculated using magnetic models.

Create Your VirES Account:

1. **Sign up:** Visit https://vires.services/oauth/accounts/signup/
   - Use your institutional email if possible
   - Create a secure password
   - Verify your email address
2. **Log in:** Go to https://vires.services/
3. **Get your access token:**
   - Follow the detailed instructions at https://viresclient.readthedocs.io/en/latest/access_token.html
   - Your token is like a password that allows MagGeo to download data for you
   - **Important:** Keep this token secure and don't share it
4. **Configure MagGeo with your token:**

```
# Replace YOUR_TOKEN_HERE with your actual token
viresclient set_token --token YOUR_TOKEN_HERE
```

Testing Your Installation

To verify everything is working:

1. Make sure you're in the MagGeo directory and the environment is activated
2. Try running a simple test (specific commands will depend on MagGeo's documentation)

Troubleshooting Common Issues

Windows-Specific Issues

Problem: "conda is not recognized as an internal or external command"

- **Solution:** Miniconda wasn't added to PATH during installation
- **Fix:** Reinstall Miniconda and ensure you check "Add Miniconda3 to my PATH environment variable"

Problem: "Permission denied" errors

- **Solution:** Run Command Prompt as Administrator

- **Fix:** Right-click Command Prompt and select "Run as administrator"

Problem: Long path names causing issues

- **Solution:** Install MagGeo in a shorter path like `C:\MagGeo`

macOS-Specific Issues

Problem: "command not found: conda"

- **Solution:** Terminal doesn't recognize conda
- **Fix:** Close and reopen Terminal, or run `source ~/.bash_profile`

Problem: "Permission denied" during Git clone

- **Solution:** You don't have write permissions in the current directory -
- **Fix:** Navigate to your home directory first: `cd ~`

Problem: Xcode Command Line Tools installation fails

- **Solution:** Install Xcode from the App Store first, then try again

General Issues

Problem: Environment creation fails with package conflicts

- **Solution:** Clear conda cache and try again
- **Fix:** Clean your conda enviroment cache and reinstall the MagGeo environment.

```
`conda clean --all`
```

```
`conda env create --file environment.yml`
```

Problem: VirES token authentication fails

- **Solution:** Token might be incorrect or expired
- **Fix:** Generate a new token from the VirES website and reconfigure

Problem: Git clone fails with "Repository not found"

- **Solution:** Check your internet connection and the repository URL
- **Fix:** Try cloning again, or download the ZIP file directly from GitHub

Reporting an Issue

If you encounter issues not covered here:

Before Reporting an Issue:

1. Note your operating system (Windows 10/11, macOS version)
2. Copy any error messages exactly as they appear
3. Note which step in the installation process failed

How to Report an Issue:

1. Visit https://github.com/MagGeo/MagGeo-Annotation-Program/issues
2. Click "New Issue"
3. Choose "Bug report" or "Installation help"
4. Provide a clear title like "Installation fails at Step 5 on Windows 11"
5. Include:
    - Your operating system
    - The exact error message
    - What you were trying to do when the error occurred
    - Screenshots if helpful

Next Steps

Once installation is complete:

- Read the MagGeo user guide for analysis workflows
- Prepare your animal movement data according to the format requirements

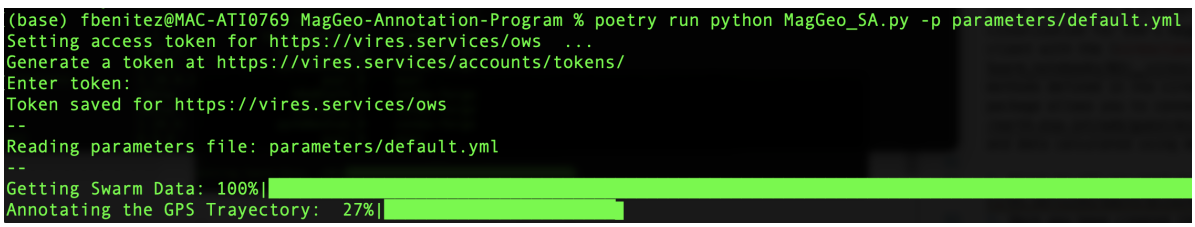
> **Tip**
>
> Remember: You only need to install MagGeo once, but you'll need to activate the environment (`conda activate MagGeoEnv`) each time you want to use it.

Run MagGeo using the sample data

MagGeo can be executed using the same terminal you have been using in the previous steps. If you want to get familiar with MagGeo and get an annotated GPS trajectory using the data we have included as an example (data folder), run the following command (replace your virES token where is requiered):

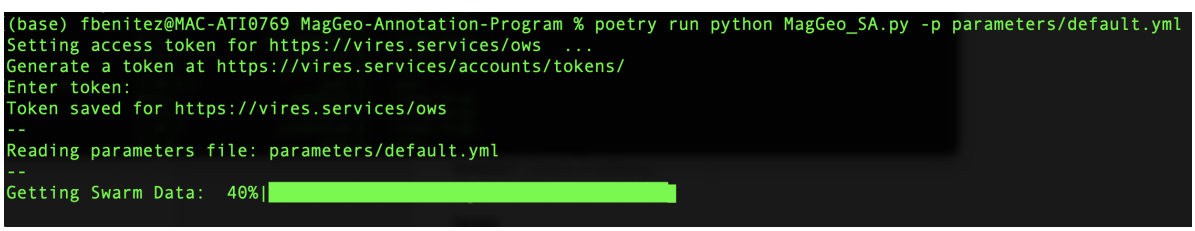
```
python MagGeo_main.py -p parameters/default.yml --token YOUR_TOKEN_HERE
```

Now MagGeo will start to download the Swarm Data.



Once the data has been downloaded, MagGeo will process it to make the annotation process (for more information about how this is done, visit our methodological paper in Movement Ecology)

The last step MagGeo does is annotating the gathered data, that would take more time depending how big is your dataset. In our example it only takes 4 seconds.



And **Congrats you got annotated data**. The results will be stored in the folder results for your futher analysis. You will find a .csv file named like **GeoMagResult_+name_of_your_csv_file_trajectory.**

Run MagGeo using your data.

If you are ready to annotate your GPS trajectories. You need to update the parameters file in MagGeo to let the program know what are the correct values of your data.

- Copy the csv file with your trajectories into the data folder.
- Open and Update the following parameters in the file `default.yml` located in parameters folder:
- `gpsfilename: "name_of_your_trajectory.cvs"` Include the name of your Input data. The GPS trajectory you need to annotate with the geomagnetic satellite data.
- `Lat: "latitude_column_name_in_your_trajectory"`
- `Long: "longitude_column_name_in_your_trajectory"`
- `DateTime: "Date_Time_column_name_in_your_trajectory"` make sure you have one column that includes Date and Time values together.
- `altitude: "altitude_column_name_in_your_trajectory"` if you do not have any altitute column, you can leave that in blank, including only `""`

Save your changes, return to the Terminal and run:

```
python MagGeo_main.py -p parameters/default.yml --token YOUR_TOKEN_HERE
```

Run MagGeo step by step, using Jupyter Notebook.

MagGeo includes a set of Jupyter Notebooks, you will find four notebooks (.ipynp) in the Notebooks folder.

In a Terminal, make sure you are using `MagGeoEnv` environment, and run:

```
jupyter notebook
```

A Jupyter Notebook dashboard will comes out in your browser locally (e.g. http://localhost:8888) then you can explore MagGeo and its content. Go to Notebooks folder and open any of the following notebook for a step by step process. You can add cells to make your own test or analysis, but be aware that any change you do at the code migth affects the correct performance of the program.

- Main Notebook : An initial and descriptive notebook where you can get detail information about MagGeo, the sample data used, background concepts and software requirements.

- Sequential Mode: Annotation Notebook applying a sequential mode. Using a traditional loop to going through the GPS track rows and process every row computing the magnetic components. Particularly useful for small datasets.
- Parallel Mode: If you have a "big" dataset (e.g. 1 million of records) you might want try the parallel mode. The parallel mode has some differences when you run the required libraries in a windows or Linux environment. We have tested **MagGeo** in a windows server environment.
- Notebook basics: If you are not familiar with Jupiter Notebooks and want to learn about the basics over how to run the notebooks before try the annotate tool. You can try this notebook to get the basics elements to run cells, read data, and plot some a basic chart.

The following image will help you to understand how the sequential and parallel mode differ, and how in parallel mode you should be able to use the full capacity of your machine. However it is quite important to identify when we need to use a parallel mode. For small datasets running **MagGeo** in Parallel mode could be even slower than the sequential mode.

Data requirements

> **!** Important
>
> Your trajectory must be in a csv format:
> There are three columns that must be included in your GPS trajectory. Make sure your GPS trajectory includes **Latitude** , **Longitude** and **timestamp**. We suggest that the Timestamp column follow the day/month/year Hour:Minute (**dd/mm/yyyy HH:MM:SS**) format, Latitude and Longitude should be in decimal degrees (WGS84). If you have a **altitude** attribute, make sure that the units are in kilometers. Other columns will be ignored. Here it is an example of how your GPS track should looks like. For this example we are reading the **BirdGPSTrajectory.csv** file. If you want to run the method using your own csv file, make sure you store your the file in the `/data` folder. For more information about the dataset we used in this example go to the Main Notebook.

Dataset used as use-case

MagGeo comes with a dataset for the use case you see in the notebook outputs. This dataset comes from the MoveBank Data Repository (https://www.datarepository.movebank.org/)[1][2].

We annotated tracking data of greater white-fronted geese (Anser albifrons), which migrate between northern Germany and the Russian Arctic. We annotated data for fifteen individuals in a single autumn migration (42 days) a total of 973 GPS location.

[1] Kölzsch A, Müskens GJDM, Kruckenberg H, Glazov P, Weinzierl R, Nolet BA, Wikelski M (2016) Towards a new understanding of migration timing: slower spring than autumn migration in geese reflects different decision rules for stopover use and departure. Oikos. doi:10.1111/oik.03121

[2] Kölzsch A, Kruckenberg H, Glazov P, Müskens GJDM, Wikelski M (2016) Data from: Towards a new understanding of migration timing: slower spring than autumn migration in geese reflects different decision rules for stopover use and departure. Movebank Data Repository. doi:10.5441/001/1.31c2v92f

Part II

The science behind

About MagGeo

Inspired by The Environmental Data Automated Track Annotation System (Env-DATA) Service a tool on Movebank, where ecologists and animal movement researchers all over the world can link movement data with global environmental datasets. Including hundreds of variables from a diverse set of data sources including the European Space Agency (ESA), the National Aeronautics and Space Administration (NASA), the US National Oceanic and Atmospheric Administration (NOAA), and others. EnvData allow researchers to annotate in space and time multiples environmental information to enrich their GPS tracks to analyze the influence of several environmental variables in the trajectory. Using the Env-DATA Track Annotation Service registered users on MoveBank are able to get environmental parameters—such as wind conditions, land use, vegetation, and snow cover—for the whole world. Using different interpolation methods users can include multiple environment variables selecting from a comprehensive list of datasets (you can browse the available dataset here).

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Key Concepts

1. Animal Navigation Fundamentals

This is not a comprehensive book related to animal navigation fundamental, however there are certain concepts you might want to recall and understand that establish the foundations of what you have built **MagGeo**.

True Navigation

Long-distance migratory navigation represents one of nature's most remarkable phenomena, consisting of two primary components that work together to enable animals to travel vast distances with inexplicable precision. The first component, is **compass orientation**, involves determining the correct direction of movement relative to environmental cues (where to go). The second component, **geographic positioning**, requires animals to know their current location at any specific time during their journey (where am I)[1].

These mechanisms together support what Ecologists term "**true navigation**" - defined as the ability to find the way to a distant, unknown location using only cues that are available locally at the animal's current position[2].

Navigation Mechanisms

Migratory animals (e.g Birds) have evolved sophisticated sensory systems that can detect and interpret multiple environmental cues for navigation purposes [@demsar2025]. These cues fall into two primary categories based on their navigational function:

For Compass Orientation: Animals use several types of compass systems to maintain consistent directional movement:

- **Solar compass**: Utilizes the Sun's position, accounting for its apparent movement across the sky and seasonal variations
- **Stellar compass**: Relies on star patterns and celestial rotation, particularly useful during nighttime migration

- **Polarized light patterns**: Detects polarization patterns in skylight that remain consistent relative to the Sun's position
- **Earth's magnetic field**: Uses the planet's magnetic field as a reliable directional reference[3]

For Geographic Positioning: To determine their specific location, animals can access several types of positional cues:

- **Earth's magnetic field**: Utilizes spatial gradients and patterns in Earth's magnetic field intensity and inclination[3]
- **Olfactory cues**: Chemical signatures that vary geographically, creating olfactory landscapes
- **Visual landmarks**: Topographical features, coastlines, and other geographical markers
- **Infrasound**: Low-frequency sound waves that can travel long distances and vary by location[1]

2. Geomagnetic Navigation

Geomagnetic navigation represents a strategy whereby animals use information derived from Earth's magnetic field for either compass orientation, geographic positioning, or both functions simultaneously[3]. This navigational mechanism has been documented across an taxonomic range, suggesting it evolved independently multiple times or represents an ancient, conserved ability.

Evidence for geomagnetic navigation exists across diverse taxa:

- **Birds**: From songbirds to seabirds, with extensive experimental evidence[1]
- **Fish**: Including both oceanic and freshwater species
- **Sea turtles**: Particularly well-documented in marine navigation [1]
- **Terrestrial mammals**: From small rodents to large ungulates[11] [12]
- **Marine mammals**: Including whales and dolphins[13] [1]

However the study of geomagnetic navigation employs two primary methodological approaches, each with distinct advantages and limitations:

Lab Experiments: Controlled studies where animals are placed in artificial magnetic field to examine how magnetic parameters influence the onset and direction of migratory behavior. Laboratory experiments provide exceptional precision and experimental control, allowing researchers to manipulate specific magnetic parameters while holding other variables constant[2] [1] [1]. However, the observed behaviors in such controlled environments differ significantly from what occurs in wild[1]. Additionally, these studies typically focus on small numbers of individuals from single species, which limits the results across multiple species[1].

Field Studies with Tracking: Modern migration research employs tracking technologies, particularly GPS tags combined with displacement experiments. These approaches provide greater ecological relevance by studying animals in their natural environments[1] [2] [21]. However, most tracking studies have been limited by their reliance on static representations of the geomagnetic field (space and temporal), failing to account for the dynamic temporal variations that may influence animal navigation[22].

3. Earth's Magnetic Field

Earth's magnetic field is a complex, dynamic system that results from the superposition of magnetic fields originating from multiple sources. Understanding this complex environmental variable is crucial for comprehending how animals use magnetic data for navigation. The total magnetic field at any location and time results from the combination of three primary components, each with different characteristics, spatial patterns, and temporal behaviors[3].

Core Field: The dominant contributor to Earth's magnetic field is the core field, generated by the geodynamo mechanism operating within Earth's outer liquid core. This field arises from the motion of electrically conductive molten iron as it circulates within the outer core, driven by thermal convection and the planet's rotation. The core field exhibits a broadly dipolar structure on large scales, resembling that of a giant bar magnet tilted approximately 11 degrees from Earth's rotational axis. However, this apparent simplicity masks considerable complexity in the detailed structure, with significant non-dipolar components that create regional variations in field strength and direction across the globe.

The core field dominates the total magnetic field at Earth's surface, typically contributing over 98% of the field strength at any location. This field undergoes gradual changes over time scales of years to decades, known as secular variation, as the circulation patterns within the outer core evolve. These long-term changes can amount to tens of nanoTesla per year at any given location, representing a slow but persistent evolution of the magnetic environment that animals experience.

Lithospheric Field: The lithospheric or crustal field originates from the magnetic properties of rocks within Earth's crust. Certain minerals, particularly magnetite and other iron-bearing compounds, became magnetized when they formed and retain this magnetization over geological time scales. The lithospheric field creates local and regional magnetic anomalies - areas

where the magnetic field strength or direction differs noticeably from what would be expected from the core field alone.

These magnetic anomalies can range from subtle variations of a few nanoTesla to dramatic variations of thousands of nanoTesla over distances of kilometers to hundreds of kilometers. The lithospheric field is considered essentially static over time periods relevant to animal lifespans, changing only over geological time scales of millennia or longer. For migratory animals, these crustal magnetic signatures provide potentially stable, location-specific magnetic features that could serve as magnetic landmarks or components of magnetic maps.

External Fields: External magnetic fields result from electrical current systems flowing in Earth's ionosphere and magnetosphere, driven by interactions between the solar wind and Earth's magnetic field. These fields exhibit the most dynamic behavior of the three components, varying on time scales from seconds to days. The external field contributions include both regular, predictable daily variations and irregular disturbances associated with space weather events.

The regular daily variations, known as Sq (solar quiet) currents, result from the regular heating of the ionosphere by solar radiation, which creates predictable current systems that vary with local time and season. These regular variations typically amount to 20-30 nanoTesla at mid-latitudes but can be significantly larger at high latitudes near the magnetic poles.

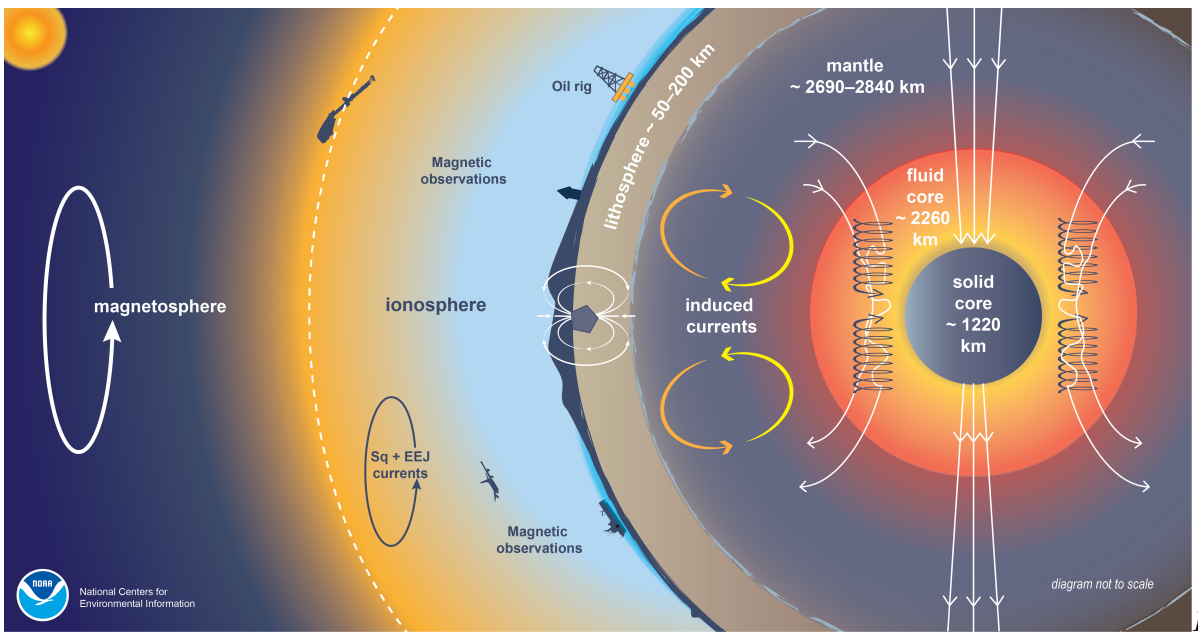


Figure 1 showing magnetic field components and coordinate system, NOOA, https://www.ncei.noaa.gov/news/HDGM

Magnetic Field Measurements and Coordinate Systems

Scientists measure Earth's magnetic field using a standardized coordinate system that allows for consistent description and comparison of magnetic field vectors worldwide. The magnetic field vector **B** is decomposed into components using the North-East-Centre (NEC) coordinate system, also known as the local geodetic coordinate system.

In this system, measurements are made relative to the local geographic reference frame at each point on Earth's surface. The North component points toward geographic north (true north), the East component points toward geographic east, and the Centre component points radially downward toward Earth's center. This coordinate system provides a consistent framework for describing magnetic field variations across the globe and through time.

From these three orthogonal components, several important magnetic field parameters are derived:

- **Intensity (F)**: The magnitude or total strength of the magnetic field vector B, representing the length of the vector in three-dimensional space
- **Inclination (I)**: The angle between the magnetic field vector B and its horizontal component H, measuring how steeply the field lines plunge into or emerge from Earth
- **Declination (D)**: The angle between the horizontal component H and geographic north, indicating how much magnetic north deviates from geographic north
- **Horizontal component (H)**: The horizontal projection of the magnetic field vector, representing the component parallel to Earth's surface

These parameters provide different types of information that animals might potentially use for navigation. Intensity varies systematically with latitude and can provide positional information, while inclination also varies predictably with latitude and could serve as a magnetic latitude indicator. Declination provides information about magnetic versus geographic north, potentially useful for compass orientation.

4. Spatial and Temporal Variation

Global Spatial Patterns

Intensity: The total magnetic field intensity shows global patterns that reflects the underlying dipolar structure of the core field. At the magnetic equators, where the field lines run horizontally, the total intensity reaches its minimum values of approximately *23,000 nanoTesla*. Moving toward higher magnetic latitudes, the field intensity gradually increases as the field lines become more vertical, reaching maximum values of approximately *62,000 nanoTesla* near the magnetic poles.

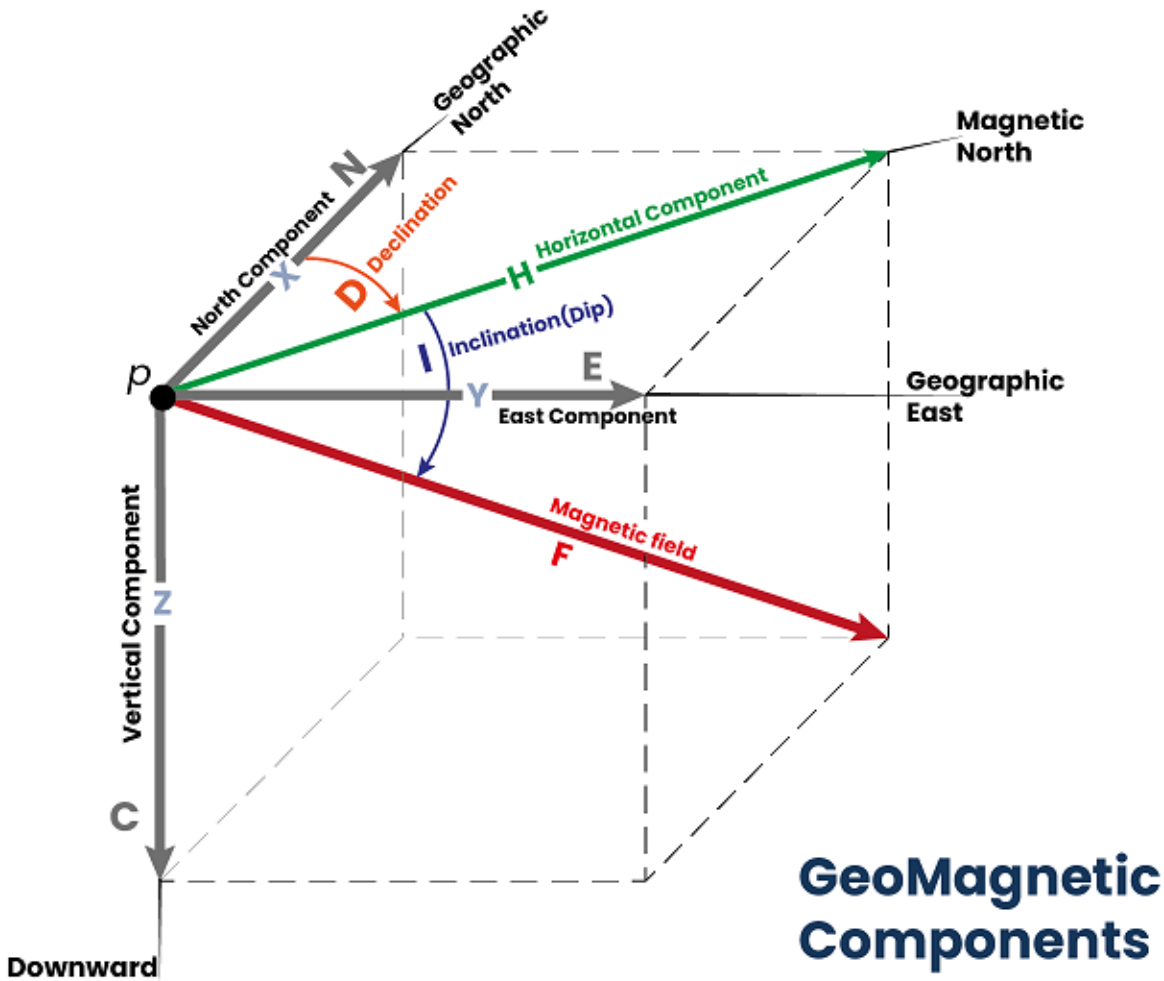
This variation in intensity with magnetic latitude creates what researchers term "isodynamic lines" - contours of equal magnetic intensity that generally run east-west but show considerable deviation from simple latitudinal bands due to non-dipolar components of the field. These intensity gradients are steep enough that animals traveling north-south distances of hundreds of kilometers would encounter measurable changes in field intensity.

Inclination: Magnetic inclination, the angle at which field lines plunge into Earth, provides perhaps the most systematic navigational cue available from the geomagnetic field. At the magnetic equator, field lines run horizontally, giving inclination values near zero degrees. Moving toward the magnetic poles, inclination increases progressively, reaching 90 degrees (vertical) at the magnetic poles themselves.

This relationship between inclination and magnetic latitude has led researchers to propose that inclination serves as a primary magnetic coordinate for animal navigation. The inclination gradient is steep enough that latitudinal movements of tens of kilometers produce detectable changes in inclination angle, potentially providing fine-scale positional information.

Declination: Magnetic declination on the other side, the deviation of magnetic north from geographic north, shows the most complex spatial patterns of the major magnetic field components. Unlike intensity and inclination, declination does not follow simple latitudinal patterns but instead shows regional variations that reflect the non-dipolar components of the core field and crustal magnetic anomalies.

In some regions, declination changes very gradually over large distances, while in other areas it can change by several degrees over distances of hundreds of kilometers. These declination patterns create distinctive regional magnetic signatures that could potentially serve as magnetic landmarks for animal navigation systems.



Temporal Variations

The temporal behavior of Earth's magnetic field spans an enormous range of time scales, from rapid variations lasting seconds to gradual changes occurring over geological time. Understanding these temporal variations is crucial for reading how animals use magnetic data for navigation.

Quiet-Time Variations: Even during geomagnetically quiet periods, Earth's magnetic field shows regular daily variations driven by solar heating of the ionosphere. During daylight hours, solar ultraviolet radiation ionizes atmospheric gases, creating electrical conductivity that allows current systems to flow in the ionosphere. These currents generate magnetic fields that add to the main field measured at Earth's surface.

The amplitude of these quiet-time variations depends strongly on latitude and local time. At mid-latitudes, typical daily variations amount to approximately 20 nanoTesla, with maximum variations occurring during local noon when ionospheric currents are strongest. At polar latitudes, where ionospheric current systems are more complex, daily variations can reach 100 nanoTesla or more.

These regular variations follow *predictable* patterns that repeat daily and show seasonal modulations. For animals using magnetic cues for navigation, these predictable variations might either be filtered out by their sensory systems or potentially used as additional timing information.

Geomagnetic Storm Disturbances: During periods of enhanced solar activity, Earth's magnetic field can experience big disturbances that far exceed normal quiet-time variations. Geomagnetic storms occur when disturbances in the solar wind create enhanced coupling between the solar wind's magnetic field and Earth's magnetosphere. These events drive intense electrical currents in the magnetosphere and ionosphere, creating magnetic field variations that can exceed *1000 nanoTesla i*n polar regions and *250 nanoTesla at mid-latitudes*[23].

The temporal characteristics of geomagnetic storms are particularly relevant for animal navigation. Storm-related magnetic field changes can occur within seconds to days, creating rapid alterations in the magnetic environment that animals experience. The magnitude of these changes often exceeds the established sensitivity thresholds of animal magnetoreceptors by factors of ten or more, suggesting that geomagnetic storms could significantly disrupt magnetic navigation systems.

Secular Variation: On longer time scales, Earth's magnetic field undergoes continuous slow changes known as secular variation, driven by evolving circulation patterns within the liquid outer core. These changes typically amount to tens of nanoTesla per year at any given location, representing a gradual evolution of the magnetic field over time scales of years to decades.

For long-lived migratory animals or populations with strong site fidelity, secular variation could potentially require periodic recalibration of magnetic navigation systems. However, the slow rate of secular variation means that it is unlikely to affect navigation during individual migratory journeys, which typically last weeks to months.

5. Geomagnetic Storms and Solar Wind

Storm-Related Disruption

The sensitivity thresholds established by neurophysiological and behavioral research have profound implications when compared to the magnitude of magnetic field variations that occur during geomagnetic storms. Solar wind disturbances can create magnetic field changes exceeding 1000 nanoTesla within minutes during major geomagnetic storms[2] , representing variations that are 5-65 times larger than established animal sensitivity thresholds.

Potential Disruption Mechanisms: Animals using magnetic intensity as a positional cue could experience significant navigation errors during geomagnetic storms. If birds or other animals rely on specific intensity values as location markers, sudden storm-related intensity changes could create the illusion that they have been displaced to entirely different geographic locations. This effect would be similar to the responses observed in virtual magnetic displacement experiments[1] [1] [1] [2] , where animals exposed to artificial magnetic fields characteristic of distant locations attempt to compensate by changing their flight direction.

Animals using magnetic inclination or declination for compass orientation might experience disruption of their directional sensing capabilities during storms. Rapid changes in field direction could cause temporary loss of magnetic compass function, potentially forcing animals to rely on alternative compass systems such as solar or stellar compasses .

Species-Specific Responses: Different animal species likely respond to geomagnetic disturbances in various ways, depending on their particular methods of using magnetic information for navigation[3]. Some species might temporarily cease migration during severe magnetic disturbances, while others might switch to alternative navigation systems. The behavioral responses probably depend on the availability of backup navigation cues and the flexibility of each species' navigation system.

6. Geomagnetic Data Sources

Terrestrial Observatory Networks

The systematic measurement of Earth's magnetic field began in the 1830s and has evolved into a sophisticated global network of ground-based observatories that provide continuous, high-precision magnetic field measurements.

INTERMAGNET Network: The International Real-time Magnetic Observatory Network (INTERMAGNET) represents the primary source of ground-based geomagnetic data, currently comprising 152 observatories distributed across the globe[3] . These observatories employ standardized instrumentation and data collection procedures to ensure consistency and comparability of measurements worldwide.

Each INTERMAGNET observatory typically operates multiple magnetometers, including absolute instruments that measure the true magnetic field vector and variometers that detect changes in the field with high temporal resolution. The observatories record magnetic field measurements continuously, typically at one-second or one-minute intervals, creating detailed time series that capture both gradual changes and rapid fluctuations.

Advantages of Terrestrial Measurements: Ground-based observatories offer several significant advantages for magnetic field research. Their calibration accuracy is exceptionally high, with measurement uncertainties typically less than 1 nanoTesla for carefully maintained instruments. The continuous operation of these stations creates long-term data records, some

extending back more than a century, that are invaluable for studying secular variation and long-term magnetic field changes.

The temporal resolution of ground-based measurements is also superior to satellite data, with one-second sampling rates that can capture rapid magnetic field fluctuations associated with geomagnetic storms and other space weather events. This high temporal resolution makes ground-based data particularly valuable for studying the detailed time evolution of magnetic disturbances.

Limitations of Ground-Based Networks: Despite their high accuracy and temporal resolution, terrestrial magnetic observatories have significant limitations for global magnetic field studies. The spatial coverage is highly irregular, with dense networks in some regions (particularly northern and western Europe) but sparse or absent coverage in others. Oceanic regions are particularly poorly covered, with very few magnetic observatories on islands and none on the open ocean.

The effective spatial coverage of each observatory is limited to approximately 1000 kilometers, beyond which local magnetic variations can differ significantly from those measured at the observatory. This limitation means that only regions with dense observatory networks provide complete spatial coverage, excluding most animal migration pathways that cross oceanic or remote continental regions.

Temporal data availability also presents challenges for real-time applications. Observatories submit their data to the central INTERMAGNET network at different times, often with delays of months to years for the highest quality, fully processed data. Some observatories occasionally cease operation due to funding or technical issues, creating gaps in temporal coverage.

Satellite-Based Measurements

Satellite missions have revolutionized geomagnetic field research by providing consistent global coverage with standardized instrumentation. Over the past 60 years, a series of dedicated magnetic field missions has complemented terrestrial measurements and enabled global-scale studies of magnetic field structure and dynamics.

Historical Satellite Missions: The evolution of satellite-based geomagnetism began with early missions such as POGO (1965-1971), which provided the first global magnetic field measurements from space. Subsequent missions including Magsat (1979-80), Ørsted (1999-present), CHAMP (2000-2010), and SAC-C (2000-present) progressively improved measurement accuracy and global coverage[3] .

Each mission contributed unique capabilities and extended the temporal baseline of satellite magnetic measurements. Magsat provided the first high-precision global magnetic field model, while Ørsted began the era of continuous, long-term satellite magnetic monitoring. CHAMP delivered exceptionally precise measurements that advanced understanding of both the main field and its temporal variations.

The Swarm Mission: The European Space Agency's Swarm mission, launched in 2013, represents the current state-of-the-art in satellite-based magnetic field measurement[3] . The mission employs three identical satellites arranged in a carefully designed constellation that provides enhanced spatial and temporal sampling of the global magnetic field.

The constellation consists of satellites Alpha and Charlie, which orbit in parallel at an altitude of 480 kilometers with a separation of approximately 150 kilometers at the equator. Satellite Bravo orbits at a higher altitude of 510 kilometers in a different orbital plane that, at present, is nearly perpendicular to the Alpha-Charlie pair. This configuration provides measurements at two different altitudes and enables separation of different magnetic field sources.

Swarm Instrumentation: Each Swarm satellite carries identical scientific instrumentation designed for precise magnetic field measurement. The primary instruments include an Absolute Scalar Magnetometer (ASM) that measures the total magnetic field intensity, a Vector Field Magnetometer (VFM) that measures the three-component magnetic field vector, and a GPS receiver for precise orbit determination.

The VFM provides magnetic field measurements at 1 Hz resolution in the North-East-Centre coordinate system, calibrated using measurements from the ASM and stellar cameras that determine satellite orientation. This combination of instruments enables precise determination of both the magnitude and direction of the magnetic field vector at satellite altitude.

Advantages of Satellite Measurements: Satellite measurements resolve many of the limitations of ground-based networks by providing consistent global coverage with standardized instrumentation. The polar-orbiting design of most magnetic field satellites ensures complete global coverage within a few days, including oceanic and remote continental regions that lack ground-based observatories.

Satellite data are typically available within days to weeks of collection, much faster than the fully processed data from many ground-based observatories. The standardized instrumentation and data processing procedures also ensure consistency in measurements across different geographic regions and time periods.

The orbital perspective of satellites enables measurement of the magnetic field above most ionospheric current systems, providing information about magnetic field sources that cannot be obtained from ground-based measurements alone. This capability is particularly valuable for studying external field variations and their global patterns.

Satellite Data Access and Formats: Swarm data are made freely available through the European Space Agency's Earth Observation Policy, with data access provided through the VirES (Virtual workspace for Earth observation Scientists) web platform . The Level 1b data products provide corrected and calibrated magnetic field measurements in standard scientific formats.

However, satellite magnetic data are provided in specialized formats such as the Common Data Format (CDF) developed by NASA [3], which are not readily accessible using standard ecological data analysis software. This technical barrier has historically limited the use of

satellite magnetic data outside the specialized geophysics community, despite the open data policies of space agencies.

Geomagnetic Indices and Activity Monitoring

Understanding and quantifying the level of geomagnetic activity is crucial for both space weather applications and studies of magnetic field effects on biological systems. Geomagnetic indices provide standardized measures of magnetic field disturbance that enable comparison of activity levels across different times and locations.

K-Index System: The K-index represents the fundamental unit for measuring local geomagnetic disturbance. Calculated every three hours at individual magnetic observatories, the K-index quantifies disturbances in the horizontal component of the magnetic field on a quasi-logarithmic scale from 0 to 9[2] . A K-index of 0 indicates perfectly calm magnetic conditions, while values of 5 or higher indicate geomagnetic storm conditions.

The K-index calculation involves comparing the observed magnetic field variations during each three-hour period to the expected quiet-day variation for that location and season. This approach accounts for the regular daily variations in the magnetic field and highlights unusual disturbances that exceed normal quiet-time levels.

Kp Index: The planetary Kp index provides a global measure of geomagnetic activity by averaging K-indices from a standardized network of magnetic observatories distributed worldwide[3] . Calculated every three hours, the Kp index ranges from 0 to 9 and serves as a proxy for the energy input from the solar wind into Earth's magnetosphere.

The Kp index is widely used in space weather applications and research studies because it provides a single number that characterizes global magnetic activity levels. Kp values of 5 or higher indicate geomagnetic storm conditions, with values above 7 representing severe storms that can have significant technological and potentially biological impacts.

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Calculation of Magnetic Components

Swarm data provide information on the earth's magnetic field at the orbit level, which is above the ionosphere, where geomagnetic field is affected by the electrical currents induced by the interaction of the solar wind and magnetosphere (see the figure above to understand the vertical contributions of the earth' magnetic field). This means that to obtain the values of the magnetic field on the Earth's surface where animals are migrating, the raw measurements from Swarm need to be corrected removing the effects from the modeled values from core, crust and magnetosphere.

We do this in three steps:

1. We download the swarm residuals for the GPS date, getting essentially the unmodelled ionospheric field contribution that we haven't captured in any of the other models.

2. Because at the ground level the core and crust contributions are strongen than the ones at the satellite altitude, We use the CHAOS model to compute the core, crust and magnetosphere contributions for the particular, latitude, longitude, datetime and altitude of each GPS point.

3. Finally we add those values modeled by CHAOS values with the Swarm residuals, computing a comprehensive magnetic values than include the stronger core and crust contributions and the unmodellled values from the satellite altitude that capture all the inoshpere field contributions. For details of this correction see Supplementary Information 1 in our main paper.

> **!** Important
>
> **CHAOS**: It is a comprehensive field model, containing the modeled contributions of the time-varying core, the static crustal field, the average time-varying magnetosphere. We show calculation of the residuals (adding this parameter `residuals=True`) which means: data - `CHAOS(Core+Static+Magnetosphere)`

The **CHAOS** model is available on the server as model variables:

- CHAOS-Core: SH degree 1-20
- CHAOS-Static: SH degree 21-110
- CHAOS-MMA-Primary: Magnetosphere external field, SH degree 1-2
- CHAOS-MMA-Secondary: Magnetosphere internal induced field, SH degree 1-2

How MagGeo Works

MagGeo is a tool that contains a set of python functions to carry out a data manipulation and mathematical processes to transform the data from Swarm Constellation,include the modelled values from CHAOS model (last version in December 2019) and annotate the interpolated values into every GPS point in a given GPS trajectory.

The following process is a set of three steps, **1. Get the Swarm data**, particularly the magnetic residuals in the NEC components. **2. Run the Spatio-temporal kernel** where the script will filter the Swarm data based on our spatial-temporal cylinder, to interpolate the magnetic components in the NEC frame for the given GPS track. Finally using the enriched GPS track, the final step will be the **3. Calculation of Magnetic Components** which use the CHAOS model to compute the magnetic components at the ground level. With the interpolated magnetic values corrected by the ionospheric contribution. The script will compute the other magnetic components including F (magnetic intensity), H (Horizontal component), Declination and Inclination. For detailed information of this process go to the Main Notebook.

Interpolation (ST-IDW) and Annotation Process

Once we have requested the data we need for each point in the GPS Track and considering we have gathered the available data from the three satellites for one day (*24 hours, every 60 seconds around 1440 Swarm measures per satellite*). Now we need to `filter` in space and time the available points to compute the magnetic values for each GPS point in its particular date and time. Therefore, before running the interpolation process we set **four** functions that will require the *latitude* and *longitude* and the *epoch* time of each GPS point to filter the Swarm points into what we called **Space-Time Cylinder**. The following picture can provide a better explanation about how the points inside the space-time cylinder are included or excluded. **Figure I**, shows how the R of the Space-Time cylinder is based on the GPS point latitude. **Figure II**, illustrate the geometric components behind the space-time cylinder where the points are filtered and included in the interpolation process. Figure B, will help you to understand how the following functions compute the required parameters **Figure III** shows the idea behind the interpolation process for each GPS point, requesting and processing the available Swarm measures by the three satellites. For more information about the time-space windows please read the full paper where we explain the details of it.

SwarmMagAnnotation function:

Working as the main function for the annotation process, this functions meets the previous functions running the space-time window filters, and the computing the ST- IDW process. This function will run a interpolation process for each GPS Point considering only the Swarm points inside the Space-Time cylinder computed by the four previous functions. The return value will be an array with the values the annotated magnetic values for the GPS point. The function will be executed inside a loop going through the GPS track. This function is the annotation process per se, and through this process at first we run the interpolation getting the magnetic values in NEC reference frame and then we compute the extra magnetic values that are useful to get a better understating of the earth's magnetic field at this particular location, date and time.

Therefore, before running the interpolation process we set **four** functions that will require the *latitude* and *longitude* and the *epoch* time of each GPS point to filter the Swarm points into what we called **Space-Time Cylinder**. The following picture can provide a better explanation about how the points inside the space-time cylinder are included or excluded. **Figure I**, shows how the R of the Space-Time cylinder is based on the GPS point latitude. **Figure II**, illustrate the geometric components behind the space-time cylinder where the points are filtered and included in the interpolation process. Figure B, will help you to understand how the following functions compute the required parameters **Figure III** shows the idea behind the interpolation process for each GPS point, requesting and processing the available Swarm measures by the three satellites. For more information about the time-space windows please read the full paper where we explain the details of it.

> **Note**
>
> Auxiliary Functions:The function ST_IDW_Process includes 4 auxiliary functions to run the spatial-temporal kernel
> distance_to_GPS function: Is the function in charge to calculate the distance between each GPS Point and the Swarm Point.
> Kradius function: Is the function in charge to compute the R (radius) value in the cylinder. The R value will be considered based on the latitude of each GPS Point.
> DistJ function: This function will calculate the d value as the hypotenuse created in the triangle created amount the locations of the GPS point, the location of the Swarm points and the radius value.
> DfTime_func function: This is a time function to selected the points in the range of a the DeltaTime - DT window. The Delta time window has been set as 4 hours for each satellite trajectory.

Part III

Notebooks

MagGeo - Sequential Mode

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Keywords | Bird migration, data fusion, Earth's magnetic field, Swarm, GPS tracking

0.1 Overview

This Jupyter Notebook will guide you through the required steps to annotate your GPS tracking data with the earth's magnetic field data from Swarm (European Space Agency). This version is called Sequential Mode, alternatively you can use Parallel Mode to take advantage of parallelized computing if required. More information about the Swarm satellites can be found in the Main Document on the MagGeo github repository. This script will use a sequential loop to run the annotation process for each GPS Point (row) from your data.

To execute the code, you can go through each cell (pressing Crtl+Enter), you will also find inner comments `##` to describe each particular step. If you are not familiar with using Jupyter Notebooks, you might want to take some time to learn how first, for example take a look at the notebook-basics.ipynb Notebook inside MagGeo.

0.2 Data requirements

Your trajectory must be in a csv format::

There are three columns that must be included in your GPS trajectory. Make sure your GPS trajectory includes **Latitude** , **Longitude** and **timestamp**. We suggest that the Timestamp column follow the day/month/year Hour:Minute (**dd/mm/yyyy HH:MM:SS**) format, Latitude and Longitude should be in decimal degrees (WGS84). Optionally an altitude column can be used providing altitude (the altitude must be in **km**). Other Columns will be ignored. Here it is an example of how your GPS track should look:

For this example we are reading the BirdGPSTrajectory.csv: file. If you want to run the method using your own csv file, make sure you store your the file in the ./data folder. For more information about the dataset we used in this example go to the Main Notebook.

0.3 Import the required python libraries

```
import datetime as dt
from datetime import timedelta
import sys, os
import pandas as pd
import numpy as np
from tqdm import tqdm
import matplotlib.pyplot as plt
from viresclient import set_token
sys.path.append("..")

import utilities
from utilities.MagGeoFunctions import getGPSData
from utilities.MagGeoFunctions import Get_Swarm_residuals
from utilities.MagGeoFunctions import ST_IDW_Process
from utilities.MagGeoFunctions import CHAOS_ground_values
```

0.4 Add your VirES web client Token

The **VirES client API**, requires a token. Before start you need to get your own VirES token. You can visit https://vires.services/ to get yours, and then add it into the next cell.

```
set_token("https://vires.services/ows", set_default=True)
```

0.5 Read the GPS track

The following steps will load the GPS track from a csv file, and set some requirements before downloading geomagnetic data from Swarm. If your csv track file doesnt not have any altitude attribute, MagGeo will use sea level as your altitude (i.e., 0 Km). **Altitude column units must be Km**

```
base_dir=os.path.dirname(os.getcwd())
temp_results_dir = os.path.join(base_dir, "temp_data")
results_dir = os.path.join(base_dir, "results")
data_dir = os.path.join(base_dir, "data")
utilities_dir = os.path.join(base_dir, "utilities")
```

```
# Make sure the csv file of your trackectory is stored in the Data folder.
# Enter the name of your GPS track csv file including the extension .csv  and press Enter (e
# Make sure you have a columnn that integrates date and time, before include in MagGeo.
# If your csv track file does not have any altitude attribute, MagGeo will use sea level as y
# i.e height (Only in KM)
gpsfilename= "BirdGPSTrajectoryTest.csv"
Lat="location-lat"
Long="location-long"
DateTime="timestamp"
altitude = "height"
```

```
# Here MagGeo is reading your CSV file, taking the Lat, Long, Date&Time and Altitutes attribu
# Setting the date and time attributes for the required format and computing the epoch colum
GPSData = getGPSData(data_dir,gpsfilename,Lat,Long,DateTime,altitude)
GPSData
```

0.6 Validate the correct amount of Swarm measures

The following loop is identifiying the time and validating if the time is less than 4:00 hours and more than 20:00 hours to bring one extra day of data. The result of this validation is written in an empty python list which will be later validated to get the unique dates. This avoids duplicate downloading of data for the same day and reduces overall computational time.

```
datestimeslist = []
for index, row in GPSData.iterrows():
    datetimerow  = row['gpsDateTime']
    daterow = row['dates']
    hourrow = row['times']
    hourrow = hourrow.strftime('%H:%M:%S')
    if hourrow < '04:00:00':
    date_bfr = daterow - (timedelta(days=1))
    datestimeslist.append(daterow)
    datestimeslist.append(date_bfr)
    if hourrow > '20:00:00':
    Date_aft = daterow + (timedelta(days=1))
    datestimeslist.append(daterow)
    datestimeslist.append(Date_aft)
    else:
    datestimeslist.append(daterow)
```

Getting a list of unique dates to download the Swarm Data

```
def uniquelistdates(list):
    x = np.array(list)
    uniquelist = np.unique(x)
    return uniquelist

uniquelist_dates = uniquelistdates(datestimeslist)
uniquelist_dates
```

0.7 Download Swarm residuals data

Once the date and time columns have been defined and the unique dates are identified the script can start the download process. Usually the data from Swarm is requested using only one satellite, however **MagGeo** will use the magnetic measures from the three satellite of the Swarm Mission (Alpha, Bravo, Charlie). Be aware satellite Charlie, got its AMS broken earlier in the mission, although the initial dates still have valid data MagGeo can use.

Be aware:: Due to the amount of dates in the demo GPS track (42 days), the time to process the sample data will take approximately 10 minutes. Unfortunatly the download process migth be a slow process, particually for the magnetic models data MagGeo requieres.

Set a connection to the VirES client and using the function Get_Swarm_residuals we will get the swarm residuals for the dates included in the previous list.

```
%%time

hours_t_day = 24 #MagGeo needs the entire Swarm data for each day of the identified day.
hours_added = dt.timedelta(hours = hours_t_day)

listdfa = []
listdfb = []
listdfc = []

for d in tqdm(uniquelist_dates, desc="Getting Swarm Data"):
    #print("Getting Swarm data for date:",d )
    startdate = dt.datetime.combine(d, dt.datetime.min.time())
    enddate = startdate + hours_added
    SwarmResidualsA,SwarmResidualsB,SwarmResidualsC = Get_Swarm_residuals(startdate, enddate
    listdfa.append(SwarmResidualsA)
    listdfb.append(SwarmResidualsB)
    listdfc.append(SwarmResidualsC)
```

Concat the previous results and temporally save the requested data locally: Integrate the previous list for all dates, into pandas dataframes. We will temporally saved the previous results, in case you need to re-run MagGeo, with the following csv files you will not need to run the download process.

```
%%time

PdSwarmRes_A = pd.concat(listdfa, join='outer', axis=0)
PdSwarmRes_A.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_A.csv'), header=True)
PdSwarmRes_B = pd.concat(listdfb, join='outer', axis=0)
PdSwarmRes_B.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_B.csv'), header=True)
PdSwarmRes_C = pd.concat(listdfc, join='outer', axis=0)
PdSwarmRes_C.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_C.csv'), header=True)

TotalSwarmRes_A = pd.read_csv(os.path.join(temp_results_dir,"TotalSwarmRes_A.csv"),low_memory
TotalSwarmRes_A['timestamp'] = pd.to_datetime(TotalSwarmRes_A['timestamp'])
TotalSwarmRes_B = pd.read_csv(os.path.join(temp_results_dir,"TotalSwarmRes_B.csv"),low_memory
TotalSwarmRes_B['timestamp'] = pd.to_datetime(TotalSwarmRes_B['timestamp'])
TotalSwarmRes_C = pd.read_csv(os.path.join(temp_results_dir,"TotalSwarmRes_C.csv"),low_memory
TotalSwarmRes_C['timestamp'] = pd.to_datetime(TotalSwarmRes_C['timestamp'])

TotalSwarmRes_A.head(10) #If you need to take a look of the Swarm Data, you can print TotalSw
```

0.8 Spatio-Temporal filter and interpolation process (ST-IDW)

Once we have requested the swarm data, now we need to `filter` in space and time the available points to compute the magnetic values (NEC frame) for each GPS point based on its particular date and time. The function ST_IDW_Process takes the GPS track and the downloaded data from swarm to filter in space and time based on the criteria defined in our method. With the swarm data filtered we interpolate (IDW) the NEC components for each GPS data point.

```
%%time
#Sequential mode, applying a traditional loop using iterrows.
if __name__ == '__main__':
    dn = [] ## List used to add all the GPS points with the annotated MAG Data. See the last
    for index, row in tqdm(GPSData.iterrows(), total=GPSData.shape[0], desc="Annotating the (
    GPSLat = row['gpsLat']
    GPSLong = row['gpsLong']
    GPSDateTime = row['gpsDateTime']
    GPSTime = row['epoch']
    GPSAltitude = row['gpsAltitude']
```

```
    #print("Process for:", index,"DateTime:",GPSDateTime)
    try:
    result=ST_IDW_Process(GPSLat,GPSLong,GPSAltitude, GPSDateTime,GPSTime, TotalSwarmRes_A, T
    dn.append(result)
    except:
    #print("Ups!.That was a bad Swarm Point, let's keep working with the next point")
    result_badPoint= {'Latitude': GPSLat, 'Longitude': GPSLong, 'Altitude':GPSAltitude, 'Date
    dn.append(result_badPoint)
    continue
```

Temporally save the ST-IDW result locally. Still MagGeo needs to run the calculation of geomagnetic components, brining the magnetic values at the altitude provided for your GPS track.

```
GPS_ResInt = pd.DataFrame(dn)
GPS_ResInt.to_csv (os.path.join(temp_results_dir,"GPS_ResInt.csv"), header=True)
GPS_ResInt
```

0.9 Compute the magnetic components at the trajectory altitute using CHAOS model

The function CHAOS_ground_values is used to run the calculation of magnetic components. This adjustment requeries the magnetic components at the trajectory altitude (or at the ground level) using CHAOS (theta, phi, radial). This process also further conducts the rotation and transformation between a geocentric earth-based reference system (CHAOS) and geodetic earth-based reference system (GPS track). Once the corrected values are calculated the non-necesary columns are removed. For more information about this process go to the Main Notebook.

```
%%time
X_obs, Y_obs, Z_obs, X_obs_internal, Y_obs_internal, Z_obs_internal =CHAOS_ground_values(util
GPS_ResInt['N'] =pd.Series(X_obs)
GPS_ResInt['E'] =pd.Series(Y_obs)
GPS_ResInt['C'] =pd.Series(Z_obs)
GPS_ResInt['N_Obs'] =pd.Series(X_obs_internal)
GPS_ResInt['E_Obs'] =pd.Series(Y_obs_internal)
GPS_ResInt['C_Obs'] =pd.Series(Z_obs_internal)

GPS_ResInt.drop(columns=['N_res', 'E_res','C_res'], inplace=True)
GPS_ResInt
```

0.10 The final result

With the NEC components for each GPS Track point, it is possible to compute the aditional magnetic components. For more information about the magnetic components and their relevance go to the main paper or notebook.

> **Tip**
>
> The annotated dataframe will include the following attributes: If you need more information about how the geomagnetic components are described go to the main MagGeo Notebook:
>
> - Latitude: from the GPS Track
> - Longitude: from the GPS Track
> - Timestamp: from the GPS Track.
> - Magnetic Field Intensity: mapped as Fgps in nanoTeslas (nT).
> - N (Northwards) component: mapped as N in nanoTeslas (nT).
> - E (Eastwards) component: mapped as E. in nanoteslas (nT).
> - C (Downwards or Center): component mapped as C in nanoTeslas (nT).
> - Horizontal component: mapped as H in nanoTeslas (nT).
> - Magnetic Inclination : mapped as I in degrees.
> - Magnetic Declination or dip angle: mapped as D in degrees.
> - Kp Index: mapped as kp.
> - Total Points: as the amount of Swarm points included in the ST-IDW process from the three satellites.
> - Minimum Distance: mapped as MinDist, representing the minimum distance from a Swarm points and each GPS point location.
> - Average Distance: mapped as AvDist, representing the average distance between the Swarm points and the GPS point location.

```
%%time
# Having Intepolated and weighted the magnetic values, we can compute the other magnectic co
GPS_ResInt['H'] = np.sqrt((GPS_ResInt['N']**2)+(GPS_ResInt['E']**2))
#check the arcgtan in python., From arctan2 is saver.
DgpsRad = np.arctan2(GPS_ResInt['E'],GPS_ResInt['N'])
GPS_ResInt['D'] = np.degrees(DgpsRad)
IgpsRad = np.arctan2(GPS_ResInt['C'],GPS_ResInt['H'])
GPS_ResInt['I'] = np.degrees(IgpsRad)
GPS_ResInt['F'] = np.sqrt((GPS_ResInt['N']**2)+(GPS_ResInt['E']**2)+(GPS_ResInt['C']**2))
GPS_ResInt
```

The previous dataframe (GPS_ResInt), MagGeo has computed the geomagnetic components

for each locations and time of your CSV trajectory. Now we will finish up combining the original atributes from your CSV with the annotated results from MagGeo.

```
%%time
originalGPSTrack=pd.read_csv(os.path.join(data_dir,gpsfilename))
MagGeoResult = pd.concat([originalGPSTrack, GPS_ResInt], axis=1)
#Drop duplicated columns. Latitude, Longitued, and DateTime will not be part of the final res
MagGeoResult.drop(columns=['Latitude', 'Longitude', 'DateTime'], inplace=True)
MagGeoResult
```

0.11 Export the final results to a CSV file

```
%%time
#Exporting the CSV file
outputfile ="GeoMagResult_"+gpsfilename
export_csv = MagGeoResult.to_csv (os.path.join(results_dir,outputfile), index = None, header=
```

0.12 Validate the results (Optional)

To validate the results we plot the `Fgps`column.

```
## Creating a copy of the results and setting the Datetime Column as dataframe index.
ValidateDF = GPS_ResInt.copy()
ValidateDF.set_index("DateTime", inplace=True)
## Plotting the F column.
hist = ValidateDF.hist(column='F')
plt.title('F distribution')
plt.xlabel('F in nT')
plt.ylabel('# of measurements')
```

0.13 Map the GPS Track using the annotated Magnetic Values (Optional)

Now we are going to plot the annotated GPS track stored into the MagDataFinal dataframe to see the different magnetic components in a map to have a better prespective of the impact of the earth magnetic field.

```
ValidateDF.plot(kind="scatter", x="Latitude", y="Longitude",
    label="Magnetic Intensity in nT",
    c="F", cmap=plt.get_cmap("gist_rainbow"),
    colorbar=True, alpha=0.4, figsize=(10,7),
    sharex=False #This is only needed to get the x-axis label working due to a current bug in

)

plt.ylabel("Longitude", fontsize=12)
plt.xlabel("Latitude", fontsize=12)
plt.legend(fontsize=12)
plt.show()
```

```
import geopandas
import geoplot
import hvplot.pandas
gdf = geopandas.GeoDataFrame(ValidateDF, geometry=geopandas.points_from_xy(ValidateDF.Longitu
gdf.head()
```

```
gdf.hvplot(title=f'Annotated trajectory using MagGeo - F GeoMag Intensity',
   geo=True,
   c='F',
   tiles='CartoLight',
   frame_width=700,
   frame_height=500)
```

```
gdf.hvplot(title=f'Annotated trajectory using MagGeo - I Inclination',
   geo=True,
   tiles='CartoLight',
   c='I',
   cmap='Viridis',
   frame_width=700,
   frame_height=500)
```

```
world = geopandas.read_file(geopandas.datasets.get_path('naturalearth_lowres'))

ax = world.plot(color='white', edgecolor='gray', figsize = (18,8))

minx, miny, maxx, maxy = gdf.total_bounds
ax.set_xlim(minx, maxx)
ax.set_ylim(miny, maxy)
```

```
gdf.plot(ax=ax, column='F', legend=True,
     legend_kwds={'label': "Magnetic Intensity in nT",
      'orientation': "horizontal"})
plt.ylabel("Longitude", fontsize=12)
plt.xlabel("Latitude", fontsize=12)

plt.show()
```

```
fig, (ax1, ax2) = plt.subplots(ncols=2, figsize = (18,8))

ax1 = world.plot(ax=ax1, color='white', edgecolor='black')
xlim = ([gdf.total_bounds[0],  gdf.total_bounds[2]])
ylim = ([gdf.total_bounds[1],  gdf.total_bounds[3]])
ax1.set_xlim(xlim)
ax1.set_ylim(ylim)


gdf.plot(ax=ax1, column='F', legend=True,
     legend_kwds={'label': "Magnetic Intensity in nT",
      'orientation': "horizontal"})
plt.ylabel("Longitude", fontsize=9)
plt.xlabel("Latitude", fontsize=9)
ax1.set_title('Magnetic Intensity - F')
ax1.set_xlabel('Latitude')
ax1.set_ylabel('Longitude')


ax2 = world.plot( ax=ax2, color='white', edgecolor='black')
xlim = ([gdf.total_bounds[0],  gdf.total_bounds[2]])
ylim = ([gdf.total_bounds[1],  gdf.total_bounds[3]])
ax2.set_xlim(xlim)
ax2.set_ylim(ylim)


# We can now plot our ``GeoDataFrame``.
gdf.plot(ax=ax2, column='I', legend=True, cmap='Spectral',
     legend_kwds={'label': " Inclination in Degrees",
      'orientation': "horizontal"})
ax2.set_title('Inclination - I')
ax2.set_xlabel('Latitude')
ax2.set_ylabel('Longitude')
```

MagGeo - Parallel Mode

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0.14 Overview

This Jupyter Notebook will guide you through the required steps to annotate your GPS tracking data with the earth's magnetic field data from Swarm (European Space Agency). This version is called Parallel Mode to take advantage of parallelized computing to process big datasets.

To execute the code, you can go through each cell (pressing Crtl+Enter), you will also find inner comments `##` to describe each particular step. If you are not familiar with Jupyter Notebook, you migth want to take some time to learn how to use it first, for example take a look at the notebook-basics.ipynb Notebook inside MagGeo.

For parallel processing, there are some considerations to make:

1. Linux and Windows environments have some differences. In windows we need to separate the functions and store them separately, then import them into a `main` function.
2. Defining what part of the process is CPU bound and what part is I/O bound: Identify what parts of the program are I/O bound (writing or reading from the disk or network) and what part par CPU bound (Processing capacity). To take advantage of our CPU capacity we need to identify the process where the CPU is actually doing the main Tasks.

0.15 Data requirements

Your trajectory must be in a csv format:

There are three columns that must be included in your GPS trajectory. Make sure your GPS trajectory includes **Latitude** , **Longitude** and **timestamp**. We suggest that the Timestamp

column follow the day/month/year Hour:Minute (**dd/mm/yyyy HH:MM:SS**) format, Latitude and Longitude should be in decimal degrees (WGS84). Optionally an altitude column can be used providing altitude (the altitude must be in **km**). Other Columns will be ignored. Here it is an example of how your GPS track should look:

For this example we are reading the BirdGPSTrajectory.csv file. If you want to run the method using your own csv file, make sure you store your the file in the ./data folder. For more information about the dataset we used in this example go to the Main Notebook.

0.16 Import the requeried libraries

```
import datetime as dt
from datetime import timedelta
import sys, os
import pandas as pd
import numpy as np
from tqdm import tqdm
from pathlib import Path
import matplotlib.pyplot as plt

from viresclient import set_token
sys.path.append("..")
import utilities
from utilities.MagGeoFunctions import getGPSData
from utilities.MagGeoFunctions import Get_Swarm_residuals
```

0.17 Add your VirES web client Token

The **VirES client API**, requires a token. Before start you need to get your own VirES token. You can visit https://vires.services/ to get yours, and then add it into the next cell.

```
set_token("https://vires.services/ows", set_default=True)
```

0.18 Reading the GPS track

The following steps will load the GPS track from a csv file, and set some requirements before download the data from Swarm. Importing the GPS track. You can note that there is a folder to store the CSV file. Using `os.getcwd()` you can validate where the file is located.

```
base_dir=os.path.dirname(os.getcwd())
temp_results_dir = os.path.join(base_dir, "temp_data")
results_dir = os.path.join(base_dir, "results")
data_dir = os.path.join(base_dir, "data")
```

```
#Make sure the csv file of your trajectory is stored in the Data folder.
#Enter the name of your GPS track csv file including the extension .csv  and press Enter (e.g
# Make sure you have a column that integrates date and time, before include in MagGeo.
#If your csv track file does not have any altitude attribute, MagGeo will use sea level as yo
# i.e height (Only in KM)

gpsfilename= "BirdGPSTrajectoryTest.csv"
Lat="location-lat"
Long="location-long"
DateTime="timestamp"
altitude = "height"
```

```
# Here MagGeo is reading your CSV file, taking the Lat, Long, Date&Time and Altitude attribut
# Setting the date and time attributes for the required format and computing the epoch colum
GPSData = getGPSData(data_dir,gpsfilename,Lat,Long,DateTime,altitude)
GPSData
```

Setting the date and time attributes for the requerided format and computing the epoch column. Values like Maximum and Minimun Date and time are also calculated.

0.19 Validate the right amount of Swarm measures

The following loop is identifiying the time and validating if the time is less than 4:00 hours and more than 20:00 hours to bring one extra day of data. The result of this validation is written in a empty python list which will be later validated to get the unique dates avoing to download data for the same day and reducing the the downloand time process.

```
%%time
datestimeslist = []
for index, row in GPSData.iterrows():
    datetimerow  = row['gpsDateTime']
    daterow = row['dates']
    hourrow = row['times']
    hourrow = hourrow.strftime('%H:%M:%S')
    if hourrow < '04:00:00':
```

```
        date_bfr = daterow - (timedelta(days=1))
        datestimeslist.append(daterow)
        datestimeslist.append(date_bfr)
    if hourrow > '20:00:00':
        Date_aft = daterow + (timedelta(days=1))
        datestimeslist.append(daterow)
        datestimeslist.append(Date_aft)
    else:
        datestimeslist.append(daterow)
```

Getting a list of unique dates, to being used to download the Swarm Data

```
%%time
def uniquelistdates(list):
    x = np.array(list)
    uniquelist = np.unique(x)
    return uniquelist

uniquelist_dates = uniquelistdates(datestimeslist)
uniquelist_dates
```

0.20 Download Swarm residuals data

Once the date and time columns have been defined, and the unique dates were identified the script can start the download process. Usually the data from Swarm is requested using only one satellite, however **MagGeo** will use the magnetic measures from the three satellite of the Swarm Mission.

Be aware: Due to the amount of dates the GPS track has (42 days) to request and compute the residuals, the time to process the sample data will take approximately 10 minutes.

Set a connection to the VirES client and using the function Get_Swarm_residuals we will get the swarm residuals for the dates included in the previous list.

```
%%time

hours_t_day = 24
hours_added = dt.timedelta(hours = hours_t_day)

listdfa = []
listdfb = []
```

```
listdfc = []

for d in tqdm(uniquelist_dates, desc="Getting Swarm Data"):
    #print("Getting Swarm data for date:",d )
    startdate = dt.datetime.combine(d, dt.datetime.min.time())
    enddate = startdate + hours_added
    SwarmResidualsA,SwarmResidualsB,SwarmResidualsC = Get_Swarm_residuals(startdate, enddate
    listdfa.append(SwarmResidualsA)
    listdfb.append(SwarmResidualsB)
    listdfc.append(SwarmResidualsC)
```

Concat the previous results and temporally save the requested data locally: Integrate the previous list for all dates, into pandas dataframes. We will temporally saved the previous results, in case you need to re-run MagGeo, with the following csv files you will not need to run the download process.

```
%%time
TotalSwarmRes_A = pd.concat(listdfa, join='outer', axis=0)
TotalSwarmRes_A.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_A.csv'), header=True)
TotalSwarmRes_B = pd.concat(listdfb, join='outer', axis=0)
TotalSwarmRes_B.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_B.csv'), header=True)
TotalSwarmRes_C = pd.concat(listdfc, join='outer', axis=0)
TotalSwarmRes_C.to_csv (os.path.join(temp_results_dir,'TotalSwarmRes_C.csv'), header=True)
TotalSwarmRes_A #If you need to take a look of the Swarm Data, you can print TotalSwarmRes_B
```

0.21 Set the number of processes, and split the dataframe (GPSData) into chunks

We can set the number or processess we need to dedicate for the multiprocessing mode, of course that also depends on the number of cores the machine you are using to run **MagGeo**. You can use `multiprocessing.cpu_count()` to set the number of processes as the the number of cores your machine has. Beside that we will also to split the GPS track into chucks to dedicate each core for each chuck. For more information take a look at the Home Notebook.

```
import multiprocessing
import sklearn
from multiprocessing import Pool

NumCores = multiprocessing.cpu_count()
```

```
df_chunks = np.array_split(GPSData,NumCores)
df_chunks
```

0.22 Spatio-Temporal filter and Interpolation process (ST-IDW)

Once we have requested the swarm data, now we need to `filter` in space and time the available points to compute the magnetic values (NEC frame) for each GPS point based on its particular date and time. The function ST_IDW_Process imported in the row_handler, takes the GPS track and the downloaded data from swarm to filter in space and time based on the criteria defined in our method. With the swarm data filtered we interpolated (IDW) the NEC components for each GPS data point, based on the latitude, date, time and number of Swarm points filtered.

The function CHAOS_ground_values, inside the MagGeoFunctions file, is used to run the **Calculation of magnetic components**. This calculation requeries the magnetic components at the trajectory altitude (or at the ground level) using CHAOS (theta, phi, radial). This process include a rotation and transformation between a geocentric frame (CHAOS) and geodetic frame (GPS track). Once the corrected values are calculated, are included in the GPS track, and the non-necesary columns are removed. For more information about this process go to the Main Notebook.

0.22.1 Run the (ST-IDW) process in parallel mode

Although the next cell seems to run a small `main` function. What is happening is a call for several functions running at same time for several cores. Initially we set a pool of processes. Using the `pool` class we will distribute the assigned function among the data chucks we created. Every data chunk will be like a subset of the entire GPS track. So we need to iterate among data chunk. And inside every data chunk we need to identify the `datetime`, `epoch`, `altitude`, `latitude` and `longitude` of each row to run the interpolation & annotation process using the Swarm data we have filtered and stored in the previous steps.

The function in charge to distribute the required function (`row_handler`) among the data chunks is the map function from the `pool` class.

`row_handler.py` is an interows iteration to get the required parameter for the `ST_IDW_Process` function.

Auxiliary Functions:

ST_IDW_Process function: This is the main function in charge to read the Swarm Data already filtered, and then import DfTime_func, distance_to_GPS, Kradius , DistJ functions to compute the spatial-time cylinder and the annotation process. The return of this function

is a row (dictionary) that will be appended into a python list where all the results from the different cores. The python list from every process is concatenated into a pandas dataframe in the main function having there the whole chain of the parallel process.

distance_to_GPS function: Is the function in charge to calculate the distance between each GPS Point and the Swarm Point.

Kradius function: Is the function in charge to compute the R (radius) value in the cylinder. The R value will be considered based on the latitude of each GPS Point.

DistJ function: This function will calculate the d value as the hypotenuse created in the triangle created amount the locations of the GPS point, the location of the Swarm points and the radius value.

DfTime_func function: This is a time function to selected the points in the range of a the DeltaTime - DT window. The Delta time window has been set as 4 hours for each satellite trajectory.

CHAOS_ground_values function: This is the calculation of geomagnetic components function to get the CHAOS magnetic values and process the Nres,Eres,Cres values and transform them into the N,E,C values at the GPS altitude.

```
%%time
from functools import partial
from utilities.row_handler import row_handler

if __name__ == '__main__':
    with multiprocessing.Pool(NumCores) as pool:
        GeoMagParallelResult = pd.concat(pool.map(partial(row_handler),df_chunks), ignore_in
```

With the Parallel mode the Annotation process takes about 12 seconds to complete (We had tested the parallel process in a windows server machine with 12 cores, see the image bellow). With the same GPS track in the sequetial mode the process is complete in about 2 minutes. In the image bellow you can see how the machine create several python processes and all cores (full CPU capacity) is taken.

Multiprocessing:

is even more powerfull when you have to process a big amount of data (e.g. 2 millons of points). Although here is making a notable improvement if you have to process a big dataset the parallelization makes even more sense.

Be aware that there is no output cell in here, you can follow the parallelization progress in the Anaconda Prompt.

0.23 The final result

With the NEC components for each GPS Track point, it is possible to compute the aditional magnetic components. For more information about the magnetic components and their relevance go to the main paper or notebook.

```
<strong>  The annotated dataframe will include the following attributes:</strong> If you need
<ul>
  <li><strong>Latitude</strong> from the GPS Track.</li>
  <li><strong>Longitude</strong> from the GPS Track.</li>
  <li><strong>Timestamp</strong> from the GPS Track.</li>
  <li><strong>Magnetic Field Intensity</strong>  mapped as Fgps in nanoTeslas (nT).</li>
  <li><strong>N (Northwards) component</strong> mapped as N in nanoTeslas (nT).</li>
  <li><strong>E (Eastwards) component</strong> mapped as E. in nanoteslas (nT).</li>
  <li><strong>C (Downwards or Center)</strong> component mapped as C in nanoTeslas (nT).</li
  <li><strong>Horizontal component</strong> mapped as H in nanoTeslas (nT).</li>
  <li><strong>Magnetic Inclination </strong> mapped as I in degrees.</li>
  <li><strong>Magnetic Declination or dip angle</strong> mapped as D in degrees</li>
  <li><strong>Kp Index</strong> mapped as kp</li>
  <li><strong>Total Points</strong> as the amount of Swarm messuares included in the ST-IDW p
  <li><strong>Minimum Distance</strong> mapped as MinDist, representing the minimum distance
  <li><strong>Average Distance</strong> mapped as AvDist, representing the average distance a
</ul>
```

```
#14. Having Intepolated and weigth magnetic values, we can compute the other magnectic compon
GeoMagParallelResult['H'] = np.sqrt((GeoMagParallelResult['N']**2)+(GeoMagParallelResult['E']
#check the arcgtan in python., From arctan2 is saver.
DgpsRad = np.arctan2(GeoMagParallelResult['E'],GeoMagParallelResult['N'])
GeoMagParallelResult['D'] = np.degrees(DgpsRad)
IgpsRad = np.arctan2(GeoMagParallelResult['C'],GeoMagParallelResult['H'])
GeoMagParallelResult['I'] = np.degrees(IgpsRad)
GeoMagParallelResult['F'] = np.sqrt((GeoMagParallelResult['N']**2)+(GeoMagParallelResult['E']
GeoMagParallelResult
```

The previous dataframe (GPS_ResInt), MagGeo has computed the geomagnetic components for each locations and time of your CSV trajectory. Now we will finish up combining the original atributes from your CSV with the annotated results from MagGeo.

```
%%time

originalGPSTrack=pd.read_csv(os.path.join(data_dir,gpsfilename))
```

```
MagGeoResult = pd.concat([originalGPSTrack, GeoMagParallelResult], axis=1)
#Drop duplicated columns. Latitude, Longitued, and DateTime will not be part of the final res
MagGeoResult.drop(columns=['Latitude', 'Longitude', 'DateTime'], inplace=True)
MagGeoResult
```

0.24 Export the final results to a CSV file

```
%%time
#Exporting the CSV file
outputfile ="GeoMagResult_"+gpsfilename
export_csv = MagGeoResult.to_csv (os.path.join(results_dir,outputfile), index = None, header=
```

0.25 Validate the results (optional)

To validate the results we plot the Fcolumn.

```
## Creating a copy of the results and setting the Datetime Column as dataframe index.
ValidateDF = GeoMagParallelResult.copy()
ValidateDF.set_index("DateTime", inplace=True)
## Plotting the F column.
hist = ValidateDF.hist(column='F')
plt.title('F distribution')
plt.xlabel('F in nT')
plt.ylabel('# of measurements')
```

0.26 Mapping the GPS Track using the annotated Magnetic Values (optional)

Now we are going to plot the annotated GPS track stored into the MagDataFinal dataframe to see how the different magnetic components in a map to have a better prespective of the impact of the earth magnetic field.

```
ValidateDF.plot(kind="scatter", x="Latitude", y="Longitude",
    label="Magnetic Intensity in nT",
    c="F", cmap=plt.get_cmap("gist_rainbow"),
    colorbar=True, alpha=0.4, figsize=(10,7),
```

```
    sharex=False #This is only needed to get the x-axis label working due to a current bug in
)

plt.ylabel("Longitude", fontsize=12)
plt.xlabel("Latitude", fontsize=10)
plt.legend(fontsize=12)
plt.show()
```

```
import geopandas
import geoplot
import hvplot.pandas
gdf = geopandas.GeoDataFrame(ValidateDF, geometry=geopandas.points_from_xy(ValidateDF.Longitu
gdf.head()
```

```
gdf.hvplot(title=f'Annotated trajectory using MagGeo - F GeoMag Intensity',
           geo=True,
           c='F',
           tiles='CartoLight',
           frame_width=700,
           frame_height=500)
```

```
gdf.hvplot(title=f'Annotated trajectory using MagGeo - I Inclination',
           geo=True,
           tiles='CartoLight',
           c='I',
           cmap='Viridis',
           frame_width=700,
           frame_height=500)
```

```
world = geopandas.read_file(geopandas.datasets.get_path('naturalearth_lowres'))
ax = world.plot(color='white', edgecolor='black', figsize = (12,6))

minx, miny, maxx, maxy = gdf.total_bounds
ax.set_xlim(minx, maxx)
ax.set_ylim(miny, maxy)

# We can now plot our ``GeoDataFrame``.
gdf.plot(ax=ax, column='F', legend=True,
         legend_kwds={'label': "Magnetic Intensity in nT",
                      'orientation': "horizontal"})
plt.ylabel("Longitude", fontsize=9)
```

```
plt.xlabel("Latitude", fontsize=9)

plt.show()
```

```
fig, (ax1, ax2) = plt.subplots(ncols=2, figsize = (15,6))

ax1 = world.plot(ax=ax1, color='white', edgecolor='black')
xlim = ([gdf.total_bounds[0],  gdf.total_bounds[2]])
ylim = ([gdf.total_bounds[1],  gdf.total_bounds[3]])
ax1.set_xlim(xlim)
ax1.set_ylim(ylim)


gdf.plot(ax=ax1, column='F', legend=True,
         legend_kwds={'label': "Magnetic Intensity in nT",
                      'orientation': "horizontal"})
plt.ylabel("Longitude", fontsize=9)
plt.xlabel("Latitude", fontsize=9)
ax1.set_title('Magnetic Intensity - F')
ax1.set_xlabel('Latitude')
ax1.set_ylabel('Longitude')


ax2 = world.plot( ax=ax2, color='white', edgecolor='black')
xlim = ([gdf.total_bounds[0],  gdf.total_bounds[2]])
ylim = ([gdf.total_bounds[1],  gdf.total_bounds[3]])
ax2.set_xlim(xlim)
ax2.set_ylim(ylim)


# We can now plot our ``GeoDataFrame``.
gdf.plot(ax=ax2, column='D', legend=True, cmap='Spectral',
         legend_kwds={'label': " Declination in Degrees",
                      'orientation': "horizontal"})
ax2.set_title('Declination - D')
ax2.set_xlabel('Latitude')
ax2.set_ylabel('Longitude')
```

Troubleshooting Guide

Before Seeking Help

You should complete this checklist:

- ☐ Miniconda is installed and `conda --version` works
- ☐ Environment was created without errors
- ☐ Environment shows `(maggeo)` when activated
- ☐ Verification script runs successfully
- ☐ Jupyter Lab starts without errors

Standardized Error Reporting

When reporting problems, always include:

```
# Run these commands and include output
conda info
conda list geopandas
python --version
jupyter --version
```

Issue 1: "conda: command not found"

This is the most common issue for beginners.

Windows Solution:

```
# Option 1: Use the correct command prompt
# Search for "Anaconda Prompt" in Start Menu if available
# Or reinstall Miniconda ensuring PATH is added

# Option 2: Manually add to PATH
```

```
set PATH=%PATH%;C:\Users\%USERNAME%\miniconda3\Scripts
set PATH=%PATH%;C:\Users\%USERNAME%\miniconda3
```

macOS Solution:

```
# Add to PATH temporarily
export PATH="$HOME/miniconda3/bin:$PATH"

# Add to PATH permanently
echo 'export PATH="$HOME/miniconda3/bin:$PATH"' >> ~/.bash_profile
source ~/.bash_profile

# For zsh users (macOS Catalina and later)
echo 'export PATH="$HOME/miniconda3/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
```

Issue 2: Environment Creation Fails

Common causes and solutions:

```
# Solution 1: Clean conda cache
conda clean --all

# Solution 2: Update conda first
conda update conda

# Solution 3: Try creating environment with explicit solver
conda env create -f environment.yml --solver=classic

# Solution 4: Check internet connection and try again
# Large downloads may timeout on slow connections
```

Issue 3: Different Python Versions

All students must have Python 3.10 for consistency.

```
# Check your Python version
python --version

# If incorrect, remove environment and recreate
```

```
conda env remove --name maggeo
conda env create -f environment.yml
```

Issue 4: Package Conflicts During Installation

This indicates environment file issues:

```
# Solution: Use mamba for faster, more reliable solving
conda install mamba -n base -c conda-forge
mamba env create -f environment.yml
```

Issue 5: Jupyter Lab Won't Start

Consistency check:

```
# Ensure environment is activated
conda activate maggeo

# Verify Jupyter installation
jupyter --version

# If missing, reinstall
conda install jupyter jupyterlab -c conda-forge

# Start Jupyter Lab
jupyter lab
```

Issue 6: Import Errors Despite Successful Installation

Environment activation problem:

```
# Always activate environment first
conda activate maggeo

# Check which Python you're using
which python     # macOS/Linux
where python     # Windows

# Should show path to conda environment, not system Python
```

Emergency Reinstallation

If all else fails, sometimes it is better and quicker run a complete clean installation:

```
# Remove environment
conda env remove --name maggeo

# Clean all caches
conda clean --all

# Recreate environment
conda env create -f environment.yml
```

Additional Resources

Use the following instructions as a guide for extra resources and better familiarity with working with Python. In case you want to manage your python environment, and get extra learning resources.

Environment Management - Useful Commands

```
# List all environments
conda env list

# Activate environment
conda activate maggeo

# Deactivate environment
conda deactivate

# Update all packages in environment
conda update --all

# Install additional package
conda install package-name

# Remove environment
conda env remove --name maggeo
```

Updating the Environment

```
# Update environment from file
conda env update -f environment.yml --prune
```

Exporting Your Environment

```
# Export current environment
conda env export > my-environment.yml
```